

Skin Burns YOLO

prageethm

```
import os
import cv2
import random
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import shutil

from collections import Counter

DATASET_PATH = "/kaggle/input/skin-burn-dataset"

files = os.listdir(DATASET_PATH)
image_files = sorted([f for f in files if f.endswith((".jpg", ".jpeg", ".png"))])
label_files = sorted([f for f in files if f.endswith(".txt")])
image_names = {os.path.splitext(f)[0]: f for f in image_files}
label_names = {os.path.splitext(f)[0]: f for f in label_files}

file_data = []
for name in set(image_names.keys()).union(set(label_names.keys())):
    file_data.append({
        "Image File": image_names.get(name, "Missing"),
        "Label File": label_names.get(name, "Missing"),
        "Has Image": "Yes" if name in image_names else "No",
        "Has Label": "Yes" if name in label_names else "No"
    })
df = pd.DataFrame(file_data)
print(df.head(50))
```

	Image File	Label File	Has Image	Has Label
0	img26.jpg	img26.txt	Yes	Yes
1	Missing	img1218.txt	No	Yes
2	img1210.jpg	img1210.txt	Yes	Yes
3	img1007.jpg	img1007.txt	Yes	Yes
4	img539.jpg	img539.txt	Yes	Yes
5	img533.jpg	img533.txt	Yes	Yes
6	img582.jpg	img582.txt	Yes	Yes
7	Missing	img1204.txt	No	Yes
8	img715.jpg	img715.txt	Yes	Yes
9	img1271.jpg	img1271.txt	Yes	Yes
10	img922.jpg	img922.txt	Yes	Yes
11	img83.jpg	img83.txt	Yes	Yes
12	img140.jpg	img140.txt	Yes	Yes

13	img631.jpg	img631.txt	Yes	Yes
14	img1337.jpg	img1337.txt	Yes	Yes
15	img594.jpg	img594.txt	Yes	Yes
16	img1154.jpg	img1154.txt	Yes	Yes
17	img501.jpg	img501.txt	Yes	Yes
18	img1414.jpg	img1414.txt	Yes	Yes
19	Missing	img1191.txt	No	Yes
20	Missing	img996.txt	No	Yes
21	img296.jpg	img296.txt	Yes	Yes
22	img5.jpg	img5.txt	Yes	Yes
23	img382.jpg	img382.txt	Yes	Yes
24	img844.jpg	img844.txt	Yes	Yes
25	img270.jpg	img270.txt	Yes	Yes
26	img210.jpg	img210.txt	Yes	Yes
27	img457.jpg	img457.txt	Yes	Yes
28	Missing	img1325.txt	No	Yes
29	img883.jpg	img883.txt	Yes	Yes
30	img255.jpg	img255.txt	Yes	Yes
31	img784.jpg	img784.txt	Yes	Yes
32	img1062.jpg	img1062.txt	Yes	Yes
33	img782.jpg	img782.txt	Yes	Yes
34	img703.jpg	img703.txt	Yes	Yes
35	img234.jpg	img234.txt	Yes	Yes
36	img23.jpg	img23.txt	Yes	Yes
37	img438.jpg	img438.txt	Yes	Yes
38	img507.jpg	img507.txt	Yes	Yes
39	Missing	img1208.txt	No	Yes
40	Missing	img881.txt	No	Yes
41	img1198.jpg	img1198.txt	Yes	Yes
42	img36.jpg	img36.txt	Yes	Yes
43	Missing	img979.txt	No	Yes
44	img464.jpg	img464.txt	Yes	Yes
45	img423.jpg	img423.txt	Yes	Yes
46	img1048.jpg	img1048.txt	Yes	Yes
47	img1161.jpg	img1161.txt	Yes	Yes
48	img103.jpg	img103.txt	Yes	Yes
49	img1196.jpg	img1196.txt	Yes	Yes

```
WORKING_DIR = "/kaggle/working/clean-skin-burn-dataset"
os.makedirs(WORKING_DIR, exist_ok=True)
```

```
image_files = {os.path.splitext(f)[0] for f in
os.listdir(DATASET_PATH) if f.endswith((".jpg", ".jpeg", ".png"))}
label_files = {os.path.splitext(f)[0] for f in
os.listdir(DATASET_PATH) if f.endswith(".txt")}
extra_labels = label_files - image_files
valid_labels = label_files - extra_labels
```

```
for name in valid_labels:
    image_path = os.path.join(DATASET_PATH, f"{name}.jpg")
```

```

label_path = os.path.join(DATASET_PATH, f"{name}.txt")

if os.path.exists(image_path):
    shutil.copy(image_path, WORKING_DIR)

if os.path.exists(label_path):
    shutil.copy(label_path, WORKING_DIR)

clean_data = [{"Image": f"{name}.jpg", "Label": f"{name}.txt"} for
name in valid_labels]
df_clean = pd.DataFrame(clean_data)
print("\n Cleaned Dataset Structure:")
print(df_clean.head())

df_clean.to_csv("/kaggle/working/cleaned_dataset_structure.csv",
index=False)
print(f"\n Copied {len(valid_labels)} valid images and labels to
{WORKING_DIR}")
print("Saved dataset structure as cleaned_dataset_structure.csv")

```

```

Cleaned Dataset Structure:
      Image      Label
0  img26.jpg  img26.txt
1  img1210.jpg  img1210.txt
2  img452.jpg  img452.txt
3  img1007.jpg  img1007.txt
4  img287.jpg  img287.txt

```

```

Copied 1227 valid images and labels to /kaggle/working/clean-skin-
burn-dataset
Saved dataset structure as cleaned_dataset_structure.csv

```

```

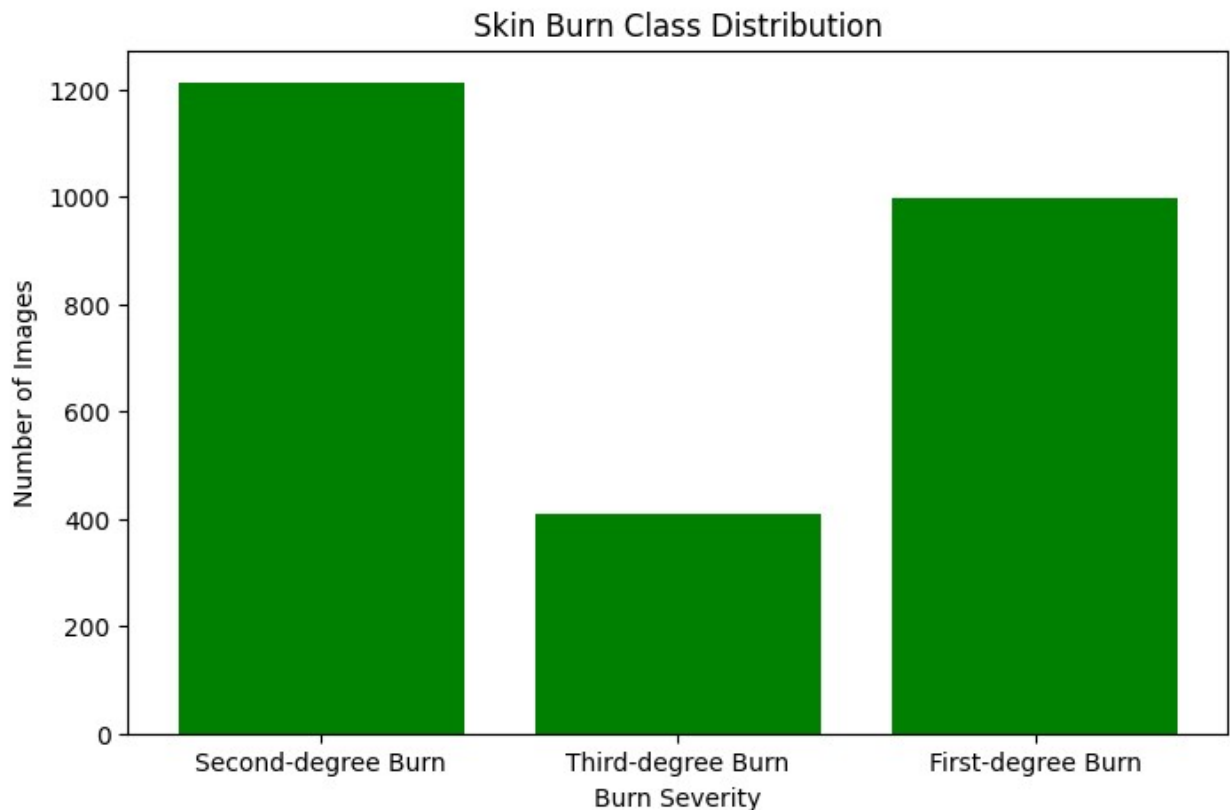
def load_yolo_labels(label_path):
    with open(label_path, "r") as file:
        lines = file.readlines()
        labels = []
        for line in lines:
            parts = line.strip().split()
            class_id = int(parts[0])
            labels.append(class_id)
        return labels

class_counts = Counter()
for label_file in os.listdir(DATASET_PATH):
    if label_file.endswith(".txt"):
        labels = load_yolo_labels(os.path.join(DATASET_PATH,
label_file))
        class_counts.update(labels)

burn_classes = {0: "First-degree Burn", 1: "Second-degree Burn", 2:
"Third-degree Burn"}

```

```
plt.figure(figsize=(8, 5))
plt.bar([burn_classes[c] for c in class_counts.keys()],
class_counts.values(), color= "green")
plt.xlabel("Burn Severity")
plt.ylabel("Number of Images")
plt.title("Skin Burn Class Distribution")
plt.show()
```



```
CLEAN_DATASET_PATH = "/kaggle/working/clean-skin-burn-dataset"
image_files = sorted([f for f in os.listdir(CLEAN_DATASET_PATH) if
f.endswith((".jpg", ".jpeg", ".png"))])
label_files = sorted([f for f in os.listdir(CLEAN_DATASET_PATH) if
f.endswith(".txt")])
num_images = len(image_files)
num_labels = len(label_files)

plt.figure(figsize=(6, 6))
plt.bar(["Images", "Labels"], [num_images, num_labels],
color="darkblue")
plt.xlabel("Category")
plt.ylabel("Count")
plt.title("Distribution of Images & Labels in Cleaned Dataset")
for i, v in enumerate([num_images, num_labels]):
```

```
plt.text(i, v + 5, str(v), ha="center", fontsize=12)
plt.show()
```

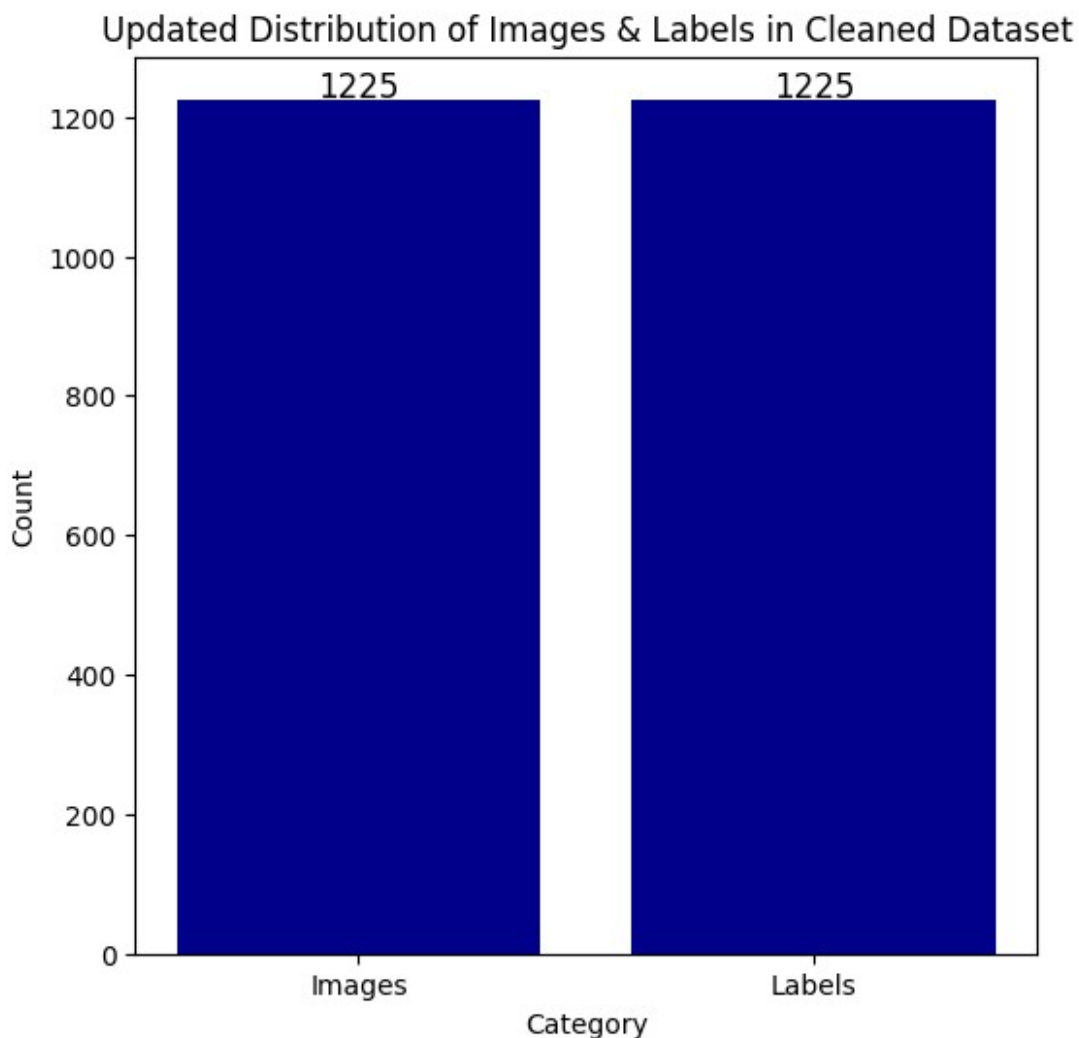


```
image_files = {os.path.splitext(f)[0] for f in
os.listdir(CLEAN_DATASET_PATH) if f.endswith((".jpg", ".jpeg",
".png"))}
label_files = {os.path.splitext(f)[0] for f in
os.listdir(CLEAN_DATASET_PATH) if f.endswith(".txt")}
extra_labels = label_files - image_files

for label in extra_labels:
    label_path = os.path.join(CLEAN_DATASET_PATH, label + ".txt")
    os.remove(label_path)
    print(f"Deleted extra label: {label_path}")
print(f"Removed {len(extra_labels)} extra labels without images!")
```

```
Deleted extra label:  
/kaggle/working/clean-skin-burn-dataset/img1424.txt  
Deleted extra label:  
/kaggle/working/clean-skin-burn-dataset/img1132.txt  
Removed 2 extra labels without images!
```

```
num_images = len([f for f in os.listdir(CLEAN_DATASET_PATH) if  
f.endswith((".jpg", ".jpeg", ".png"))])  
num_labels = len([f for f in os.listdir(CLEAN_DATASET_PATH) if  
f.endswith(".txt")])  
plt.figure(figsize=(6, 6))  
plt.bar(["Images", "Labels"], [num_images, num_labels],  
color="darkblue")  
plt.xlabel("Category")  
plt.ylabel("Count")  
plt.title("Updated Distribution of Images & Labels in Cleaned  
Dataset")  
for i, v in enumerate([num_images, num_labels]):  
    plt.text(i, v + 5, str(v), ha="center", fontsize=12)  
plt.show()
```



```
!pip install ultralytics
```

Collecting ultralytics

Downloading ultralytics-8.3.88-py3-none-any.whl.metadata (35 kB)

Requirement already satisfied: numpy<=2.1.1,>=1.23.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (1.26.4)

Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (3.7.5)

Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (4.10.0.84)

Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (11.0.0)

Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (6.0.2)

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (2.32.3)

Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (1.13.1)

Requirement already satisfied: torch>=1.8.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics)
(2.5.1+cu121)

Requirement already satisfied: torchvision>=0.9.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics)
(0.20.1+cu121)

Requirement already satisfied: tqdm>=4.64.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (4.67.1)

Requirement already satisfied: psutil in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (5.9.5)

Requirement already satisfied: py-cpuinfo in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (9.0.0)

Requirement already satisfied: pandas>=1.1.4 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (2.2.3)

Requirement already satisfied: seaborn>=0.11.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (0.12.2)

Collecting ultralytics-thop>=2.0.0 (from ultralytics)
 Downloading ultralytics_thop-2.0.14-py3-none-any.whl.metadata (9.4
kB)

Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (1.3.1)

Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (4.55.3)

Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (1.4.7)

Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (24.2)

Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (3.2.0)

Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (2.9.0.post0)

Requirement already satisfied: mkl_fft in
/usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0-
>ultralytics) (1.3.8)

Requirement already satisfied: mkl_random in
/usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0-
>ultralytics) (1.2.4)

Requirement already satisfied: mkl_umath in
/usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0-
>ultralytics) (0.1.1)

Requirement already satisfied: mkl in /usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0->ultralitics) (2025.0.1)

Requirement already satisfied: tbb4py in /usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0->ultralitics) (2022.0.0)

Requirement already satisfied: mkl-service in /usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0->ultralitics) (2.4.1)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.10/dist-packages (from pandas>=1.1.4->ultralitics) (2025.1)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.10/dist-packages (from pandas>=1.1.4->ultralitics) (2025.1)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralitics) (3.4.1)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralitics) (3.10)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralitics) (2.3.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralitics) (2025.1.31)

Requirement already satisfied: filelock in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (3.17.0)

Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (4.12.2)

Requirement already satisfied: networkx in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (3.4.2)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (3.1.4)

Requirement already satisfied: fsspec in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (2024.12.0)

Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (1.13.1)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.10/dist-packages (from sympy==1.13.1->torch>=1.8.0->ultralitics) (1.3.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.7-

```
>matplotlib>=3.3.0->ultralytics) (1.17.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.10/dist-packages (from jinja2->torch>=1.8.0-
>ultralytics) (3.0.2)
Requirement already satisfied: intel-openmp>=2024 in
/usr/local/lib/python3.10/dist-packages (from mkl-
>numpy<=2.1.1,>=1.23.0->ultralytics) (2024.2.0)
Requirement already satisfied: tbb==2022.* in
/usr/local/lib/python3.10/dist-packages (from mkl-
>numpy<=2.1.1,>=1.23.0->ultralytics) (2022.0.0)
Requirement already satisfied: tcmlib==1.* in
/usr/local/lib/python3.10/dist-packages (from tbb==2022.*->mkl-
>numpy<=2.1.1,>=1.23.0->ultralytics) (1.2.0)
Requirement already satisfied: intel-cmplr-lib-rt in
/usr/local/lib/python3.10/dist-packages (from mkl_umath-
>numpy<=2.1.1,>=1.23.0->ultralytics) (2024.2.0)
Requirement already satisfied: intel-cmplr-lib-ur==2024.2.0 in
/usr/local/lib/python3.10/dist-packages (from intel-openmp>=2024->mkl-
>numpy<=2.1.1,>=1.23.0->ultralytics) (2024.2.0)
Downloading ultralytics-8.3.88-py3-none-any.whl (932 kB)
932.9/932.9 kB 15.9 MB/s eta
0:00:00a 0:00:01
```

```
from ultralytics import YOLO
```

Creating new Ultralytics Settings v0.0.6 file ☐
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see
<https://docs.ultralytics.com/quickstart/#ultralytics-settings>.

```
YOLO_DATASET_PATH = "/kaggle/working/yolo-dataset"
os.makedirs(YOLO_DATASET_PATH, exist_ok=True)
os.makedirs(f"{YOLO_DATASET_PATH}/images/train", exist_ok=True)
os.makedirs(f"{YOLO_DATASET_PATH}/images/val", exist_ok=True)
os.makedirs(f"{YOLO_DATASET_PATH}/labels/train", exist_ok=True)
os.makedirs(f"{YOLO_DATASET_PATH}/labels/val", exist_ok=True)

image_files = sorted([f for f in os.listdir(CLEAN_DATASET_PATH) if
f.endswith((".jpg", ".jpeg", ".png"))])
label_files = sorted([f for f in os.listdir(CLEAN_DATASET_PATH) if
f.endswith(".txt")])

split_idx = int(len(image_files) * 0.8)
train_images, val_images = image_files[:split_idx],
image_files[split_idx:]

for img in train_images:
    shutil.copy(os.path.join(CLEAN_DATASET_PATH, img),
```

```

f"{YOLO_DATASET_PATH}/images/train/{img}")
    label = img.replace(".jpg", ".txt").replace(".jpeg",
".txt").replace(".png", ".txt")
    shutil.copy(os.path.join(CLEAN_DATASET_PATH, label),
f"{YOLO_DATASET_PATH}/labels/train/{label}")

for img in val_images:
    shutil.copy(os.path.join(CLEAN_DATASET_PATH, img),
f"{YOLO_DATASET_PATH}/images/val/{img}")
    label = img.replace(".jpg", ".txt").replace(".jpeg",
".txt").replace(".png", ".txt")
    shutil.copy(os.path.join(CLEAN_DATASET_PATH, label),
f"{YOLO_DATASET_PATH}/labels/val/{label}")

print(f" Dataset prepared in YOLOv8 format at {YOLO_DATASET_PATH}")

```

Dataset prepared in YOLOv8 format at /kaggle/working/yolo-dataset

```

data_yaml = f"""
path: {YOLO_DATASET_PATH}
train: images/train
val: images/val
nc: 3
names: ['First-degree Burn', 'Second-degree Burn', 'Third-degree
Burn']
"""

with open(f"{YOLO_DATASET_PATH}/data.yaml", "w") as f:
    f.write(data_yaml)

```

```
print("data.yaml file created successfully!")
```

data.yaml file created successfully!

```

model = YOLO("yolov8s.pt")
model.train(
    data=f"{YOLO_DATASET_PATH}/data.yaml",
    epochs=100,
    imgsz=640,
    patience = 20,
    batch=16,
    device="cuda"
)
print("YOLOv8 Training Completed!")

```

Downloading

<https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8s.pt> to 'yolov8s.pt'...

100%|██████████| 21.5M/21.5M [00:00<00:00, 177MB/s]

```

Ultralytics 8.3.88 □ Python-3.10.12 torch-2.5.1+cu121 CUDA:0 (Tesla
T4, 15095MiB)
engine/trainer: task=detect, mode=train, model=yolov8s.pt,
data=/kaggle/working/yolo-dataset/data.yaml, epochs=100, time=None,
patience=20, batch=16, imgsz=640, save=True, save_period=-1,
cache=False, device=cuda, workers=8, project=None, name=train,
exist_ok=False, pretrained=True, optimizer=auto, verbose=True, seed=0,
deterministic=True, single_cls=False, rect=False, cos_lr=False,
close_mosaic=10, resume=False, amp=True, fraction=1.0, profile=False,
freeze=None, multi_scale=False, overlap_mask=True, mask_ratio=4,
dropout=0.0, val=True, split=val, save_json=False, save_hybrid=False,
conf=None, iou=0.7, max_det=300, half=False, dnn=False, plots=True,
source=None, vid_stride=1, stream_buffer=False, visualize=False,
augment=False, agnostic_nms=False, classes=None, retina_masks=False,
embed=None, show=False, save_frames=False, save_txt=False,
save_conf=False, save_crop=False, show_labels=True, show_conf=True,
show_boxes=True, line_width=None, format=torchscript, keras=False,
optimize=False, int8=False, dynamic=False, simplify=True, opset=None,
workspace=None, nms=False, lr0=0.01, lrf=0.01, momentum=0.937,
weight_decay=0.0005, warmup_epochs=3.0, warmup_momentum=0.8,
warmup_bias_lr=0.1, box=7.5, cls=0.5, dfl=1.5, pose=12.0, kobj=1.0,
nbs=64, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, degrees=0.0, translate=0.1,
scale=0.5, shear=0.0, perspective=0.0, flipud=0.0, fliplr=0.5,
bgr=0.0, mosaic=1.0, mixup=0.0, copy_paste=0.0, copy_paste_mode=flip,
auto_augment=randaugument, erasing=0.4, crop_fraction=1.0, cfg=None,
tracker=botsort.yaml, save_dir=runs/detect/train
Downloading https://ultralytics.com/assets/Arial.ttf to
'/root/.config/Ultralytics/Arial.ttf'...

```

```

100%|██████████| 755k/755k [00:00<00:00, 18.0MB/s]

```

```

Overriding model.yaml nc=80 with nc=3

```

	from	n	params	module
arguments				
0	-1	1	928	ultralytics.nn.modules.conv.Conv
[3, 32, 3, 2]				
1	-1	1	18560	ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]				
2	-1	1	29056	ultralytics.nn.modules.block.C2f
[64, 64, 1, True]				
3	-1	1	73984	ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]				
4	-1	2	197632	ultralytics.nn.modules.block.C2f
[128, 128, 2, True]				
5	-1	1	295424	ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]				
6	-1	2	788480	ultralytics.nn.modules.block.C2f
[256, 256, 2, True]				
7	-1	1	1180672	ultralytics.nn.modules.conv.Conv

```

[256, 512, 3, 2]
 8          -1  1   1838080  ultralytics.nn.modules.block.C2f
[512, 512, 1, True]
 9          -1  1    656896
ultralytics.nn.modules.block.SPPF          [512, 512, 5]

10          -1  1         0
torch.nn.modules.upsampling.Upsample      [None, 2, 'nearest']

11          [-1, 6]  1         0
ultralytics.nn.modules.conv.Concat        [1]

12          -1  1   591360  ultralytics.nn.modules.block.C2f
[768, 256, 1]
13          -1  1         0
torch.nn.modules.upsampling.Upsample      [None, 2, 'nearest']

14          [-1, 4]  1         0
ultralytics.nn.modules.conv.Concat        [1]

15          -1  1   148224  ultralytics.nn.modules.block.C2f
[384, 128, 1]
16          -1  1   147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
17          [-1, 12]  1         0
ultralytics.nn.modules.conv.Concat        [1]

18          -1  1   493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
19          -1  1   590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
20          [-1, 9]  1         0
ultralytics.nn.modules.conv.Concat        [1]

21          -1  1   1969152  ultralytics.nn.modules.block.C2f
[768, 512, 1]
22          [15, 18, 21]  1   2117209
ultralytics.nn.modules.head.Detect        [3, [128, 256, 512]]

```

Model summary: 129 layers, 11,136,761 parameters, 11,136,745 gradients, 28.7 GFLOPs

Transferred 349/355 items from pretrained weights

TensorBoard: Start with 'tensorboard --logdir runs/detect/train', view at <http://localhost:6006/>

Freezing layer 'model.22.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

Downloading

<https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt> to 'yolo11n.pt'...

100%|██████████| 5.35M/5.35M [00:00<00:00, 76.5MB/s]

AMP: checks passed

train: Scanning /kaggle/working/yolo-dataset/labels/train... 980 images, 4 backgrounds, 0 corrupt: 100%|██████████| 980/980 [00:00<00:00, 998.80it/s]

train: New cache created:
/kaggle/working/yolo-dataset/labels/train.cache

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

/usr/local/lib/python3.10/dist-packages/albumentations/__init__.py:24: UserWarning: A new version of Albumentations is available: 2.0.5 (you have 1.4.20). Upgrade using: pip install -U albumentations. To disable automatic update checks, set the environment variable NO_ALBUMENTATIONS_UPDATE to 1.

check_for_updates()

val: Scanning /kaggle/working/yolo-dataset/labels/val... 245 images, 0 backgrounds, 0 corrupt: 100%|██████████| 245/245 [00:00<00:00, 934.56it/s]

val: New cache created: /kaggle/working/yolo-dataset/labels/val.cache

Plotting labels to runs/detect/train/labels.jpg...

optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automatically...

optimizer: AdamW(lr=0.001429, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)

TensorBoard: model graph visualization added

Image sizes 640 train, 640 val

Using 4 dataloader workers

Logging results to runs/detect/train

Starting training for 100 epochs...

	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
--	-------	---------	----------	----------	----------	-----------

Size	1/100	3.8G	2.296	3.619	2.26	8
------	-------	------	-------	-------	------	---

640: 100%|██████████| 62/62 [00:16<00:00, 3.68it/s]

	Class	Images	Instances	Box(P	R
--	-------	--------	-----------	-------	---

mAP50 mAP50-95): 100%|██████████| 8/8 [00:03<00:00, 2.41it/s]

		all	245	482	0.0927	0.151
0.0449	0.0125					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
------	-------	---------	----------	----------	-----------	-----------

	2/100	5.02G	2.188	3.026	2.176	17
640: 100%	[██████████] 62/62 [00:15<00:00, 4.01it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	[██████████] 8/8 [00:02<00:00, 3.94it/s]				

		all	245	482	0.0893	0.0899
0.0206	0.00609					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
------	-------	---------	----------	----------	-----------	-----------

	3/100	5.02G	2.241	3.015	2.201	25
640: 100%	[██████████] 62/62 [00:15<00:00, 4.03it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	[██████████] 8/8 [00:01<00:00, 4.49it/s]				

		all	245	482	0.0872	0.0334
0.0137	0.00343					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
------	-------	---------	----------	----------	-----------	-----------

	4/100	5.02G	2.224	3.046	2.204	17
640: 100%	[██████████] 62/62 [00:15<00:00, 3.99it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	[██████████] 8/8 [00:02<00:00, 3.93it/s]				

		all	245	482	0.0321	0.0684
0.0178	0.00575					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
------	-------	---------	----------	----------	-----------	-----------

	5/100	5.02G	2.217	3.007	2.172	15
640: 100%	[██████████] 62/62 [00:15<00:00, 3.88it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	[██████████] 8/8 [00:01<00:00, 4.30it/s]				

		all	245	482	0.0691	0.125
0.031	0.00906					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	6/100	5.02G	2.166	2.894	2.1	5
100%	62/62 [00:16<00:00, 3.80it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	8/8 [00:01<00:00, 4.20it/s]				
		all	245	482	0.147	0.137
0.0656	0.02					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	7/100	5.02G	2.152	2.84	2.075	11
100%	62/62 [00:16<00:00, 3.68it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	8/8 [00:01<00:00, 4.18it/s]				
		all	245	482	0.08	0.061
0.0389	0.0122					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	8/100	5.02G	2.103	2.796	2.043	4
100%	62/62 [00:16<00:00, 3.80it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	8/8 [00:02<00:00, 3.99it/s]				
		all	245	482	0.16	0.146
0.0779	0.0226					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	9/100	5.02G	2.135	2.786	2.069	11
100%	62/62 [00:16<00:00, 3.85it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	8/8 [00:01<00:00, 4.26it/s]				
		all	245	482	0.188	0.174
0.0952	0.029					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	10/100	5.02G	2.076	2.698	2.042	15
	100% ██████████		62/62 [00:16<00:00, 3.83it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.20it/s]			
		all	245	482	0.162	0.171
0.1	0.0314					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	11/100	5.02G	2.108	2.709	2.038	12
	100% ██████████		62/62 [00:16<00:00, 3.80it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.18it/s]			
		all	245	482	0.145	0.203
0.102	0.0316					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	12/100	5.02G	2.101	2.691	2.046	14
	100% ██████████		62/62 [00:16<00:00, 3.78it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.25it/s]			
		all	245	482	0.204	0.172
0.0842	0.0249					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	13/100	5.02G	2.089	2.657	2.009	11
	100% ██████████		62/62 [00:16<00:00, 3.82it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.31it/s]			
		all	245	482	0.141	0.171
0.0758	0.0231					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	14/100	5.02G	2.05	2.622	2.022	17
640:	100% ██████████	62/62	[00:16<00:00, 3.83it/s]			
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100% ██████████	8/8	[00:01<00:00, 4.35it/s]			
	all	245	482	0.243	0.216	
0.143	0.0428					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	15/100	5.02G	2.064	2.591	2.032	10
640:	100% ██████████	62/62	[00:16<00:00, 3.82it/s]			
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100% ██████████	8/8	[00:01<00:00, 4.24it/s]			
	all	245	482	0.194	0.204	
0.134	0.0393					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	16/100	5.02G	2.022	2.582	1.978	8
640:	100% ██████████	62/62	[00:16<00:00, 3.82it/s]			
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100% ██████████	8/8	[00:01<00:00, 4.26it/s]			
	all	245	482	0.21	0.23	
0.139	0.0445					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	17/100	5.02G	2.03	2.608	2.018	13
640:	100% ██████████	62/62	[00:16<00:00, 3.80it/s]			
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100% ██████████	8/8	[00:01<00:00, 4.19it/s]			
	all	245	482	0.267	0.2	
0.145	0.0502					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	18/100	5.02G	2.022	2.543	1.964	16
640:	100% ██████████	62/62	[00:16<00:00, 3.81it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████	8/8	[00:01<00:00, 4.27it/s]			
		all	245	482	0.204	0.227
0.124	0.0404					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	19/100	5.02G	2.004	2.519	1.966	16
640:	100% ██████████	62/62	[00:16<00:00, 3.80it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████	8/8	[00:01<00:00, 4.31it/s]			
		all	245	482	0.254	0.198
0.134	0.0462					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	20/100	5.02G	1.974	2.459	1.938	16
640:	100% ██████████	62/62	[00:16<00:00, 3.81it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████	8/8	[00:02<00:00, 3.95it/s]			
		all	245	482	0.243	0.247
0.15	0.0543					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	21/100	5.02G	1.992	2.483	1.926	15
640:	100% ██████████	62/62	[00:16<00:00, 3.80it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████	8/8	[00:01<00:00, 4.26it/s]			

		all	245	482	0.278	0.239
0.174	0.06					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	22/100	5.02G	1.99	2.449	1.923	12
640:	100% ██████████		62/62 [00:16<00:00, 3.80it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.43it/s]			

		all	245	482	0.275	0.234
0.178	0.0617					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	23/100	5.02G	1.952	2.44	1.906	13
640:	100% ██████████		62/62 [00:16<00:00, 3.81it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.28it/s]			

		all	245	482	0.293	0.241
0.16	0.0537					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	24/100	5.02G	1.946	2.422	1.92	10
640:	100% ██████████		62/62 [00:16<00:00, 3.80it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.29it/s]			

		all	245	482	0.27	0.288
0.175	0.0579					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	25/100	5.02G	1.945	2.388	1.925	17
640:	100% ██████████		62/62 [00:16<00:00, 3.80it/s]			

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%			8/8 [00:01<00:00, 4.37it/s]		
		all	245	482	0.249	0.228
0.143	0.0515					

Size	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
	26/100	5.02G	1.953	2.383	1.919	21
640:	100%		62/62 [00:16<00:00, 3.80it/s]			

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%			8/8 [00:01<00:00, 4.24it/s]		
		all	245	482	0.308	0.233
0.174	0.0622					

Size	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
	27/100	5.02G	1.911	2.324	1.883	16
640:	100%		62/62 [00:16<00:00, 3.80it/s]			

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%			8/8 [00:01<00:00, 4.27it/s]		
		all	245	482	0.264	0.245
0.162	0.0578					

Size	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
	28/100	5.02G	1.909	2.288	1.879	18
640:	100%		62/62 [00:16<00:00, 3.80it/s]			

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%			8/8 [00:01<00:00, 4.37it/s]		
		all	245	482	0.286	0.268
0.161	0.0604					

Size	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
------	-------	---------	----------	----------	----------	-----------

29/100	5.02G	1.906	2.267	1.866	13
640: 100%	██████████	62/62 [00:16<00:00, 3.80it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	8/8 [00:01<00:00, 4.36it/s]		
	all	245	482	0.304	0.261
0.171	0.0637				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
30/100	5.02G	1.899	2.307	1.884	14
640: 100%	██████████	62/62 [00:16<00:00, 3.82it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	8/8 [00:01<00:00, 4.35it/s]		
	all	245	482	0.354	0.274
0.201	0.0699				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
31/100	5.02G	1.853	2.22	1.859	12
640: 100%	██████████	62/62 [00:16<00:00, 3.78it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	8/8 [00:01<00:00, 4.28it/s]		
	all	245	482	0.267	0.223
0.15	0.059				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
32/100	5.02G	1.864	2.254	1.866	16
640: 100%	██████████	62/62 [00:16<00:00, 3.80it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	8/8 [00:02<00:00, 3.80it/s]		
	all	245	482	0.346	0.244
0.191	0.0681				

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	33/100	5.02G	1.891	2.248	1.887	10
	100% ██████████		62/62 [00:16<00:00, 3.81it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.37it/s]			
		all	245	482	0.281	0.263
0.185	0.0576					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	34/100	5.02G	1.831	2.21	1.843	12
	100% ██████████		62/62 [00:16<00:00, 3.79it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.34it/s]			
		all	245	482	0.29	0.244
0.18	0.0614					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	35/100	5.02G	1.843	2.176	1.829	15
	100% ██████████		62/62 [00:16<00:00, 3.80it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.22it/s]			
		all	245	482	0.323	0.23
0.198	0.0765					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	36/100	5.02G	1.847	2.162	1.837	15
	100% ██████████		62/62 [00:16<00:00, 3.79it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.41it/s]			
		all	245	482	0.309	0.234
0.18	0.0706					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	37/100	5.02G	1.836	2.17	1.839	12
640:	100% ██████████	62/62 [00:16<00:00, 3.80it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100% ██████████	8/8 [00:01<00:00, 4.29it/s]				
	all	245	482	0.376	0.233	
0.196	0.0613					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	38/100	5.02G	1.83	2.116	1.826	16
640:	100% ██████████	62/62 [00:16<00:00, 3.81it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100% ██████████	8/8 [00:01<00:00, 4.39it/s]				
	all	245	482	0.422	0.243	
0.226	0.0812					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	39/100	5.02G	1.812	2.076	1.825	13
640:	100% ██████████	62/62 [00:16<00:00, 3.81it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100% ██████████	8/8 [00:01<00:00, 4.33it/s]				
	all	245	482	0.41	0.261	
0.211	0.0799					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	40/100	5.02G	1.801	2.069	1.808	8
640:	100% ██████████	62/62 [00:16<00:00, 3.82it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100% ██████████	8/8 [00:01<00:00, 4.23it/s]				

		all	245	482	0.318	0.243
0.185	0.0641					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	41/100	5.02G	1.79	2.09	1.799	16
640:	100% ██████████		62/62 [00:16<00:00, 3.79it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.45it/s]			

		all	245	482	0.298	0.278
0.181	0.0567					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	42/100	5.02G	1.766	2.017	1.787	18
640:	100% ██████████		62/62 [00:16<00:00, 3.82it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.45it/s]			

		all	245	482	0.393	0.259
0.235	0.088					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	43/100	5.02G	1.789	2.005	1.783	10
640:	100% ██████████		62/62 [00:16<00:00, 3.79it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.31it/s]			

		all	245	482	0.38	0.218
0.177	0.0648					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	44/100	5.02G	1.747	1.986	1.774	9
640:	100% ██████████		62/62 [00:16<00:00, 3.80it/s]			

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%			8/8 [00:01<00:00,	4.16it/s]	
		all	245	482	0.357	0.277
0.22	0.0815					

Size	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
	45/100	5.02G	1.733	1.932	1.746	13
640:	100%		62/62 [00:16<00:00,	3.81it/s]		

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%			8/8 [00:01<00:00,	4.41it/s]	
		all	245	482	0.35	0.265
0.197	0.0642					

Size	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
	46/100	5.02G	1.738	1.932	1.758	11
640:	100%		62/62 [00:16<00:00,	3.80it/s]		

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%			8/8 [00:01<00:00,	4.27it/s]	
		all	245	482	0.375	0.252
0.187	0.0673					

Size	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
	47/100	5.02G	1.724	1.966	1.767	13
640:	100%		62/62 [00:16<00:00,	3.80it/s]		

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%			8/8 [00:01<00:00,	4.31it/s]	
		all	245	482	0.373	0.257
0.203	0.0722					

Size	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
------	-------	---------	----------	----------	----------	-----------

48/100	5.02G	1.701	1.881	1.732	14
640: 100%	██████████	62/62 [00:16<00:00, 3.81it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	8/8 [00:01<00:00, 4.47it/s]		
	all	245	482	0.397	0.257
0.208	0.0713				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
49/100	5.02G	1.69	1.881	1.717	10
640: 100%	██████████	62/62 [00:16<00:00, 3.81it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	8/8 [00:01<00:00, 4.39it/s]		
	all	245	482	0.296	0.249
0.187	0.0612				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
50/100	5.02G	1.706	1.877	1.735	9
640: 100%	██████████	62/62 [00:16<00:00, 3.80it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	8/8 [00:01<00:00, 4.34it/s]		
	all	245	482	0.324	0.311
0.221	0.0787				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
51/100	5.02G	1.695	1.846	1.725	4
640: 100%	██████████	62/62 [00:16<00:00, 3.81it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	8/8 [00:01<00:00, 4.27it/s]		
	all	245	482	0.349	0.284
0.221	0.0775				

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	52/100	5.02G	1.654	1.804	1.684	20
	100% ██████████		62/62 [00:16<00:00, 3.81it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.39it/s]			
		all	245	482	0.36	0.251
0.201	0.0685					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	53/100	5.02G	1.639	1.772	1.678	14
	100% ██████████		62/62 [00:16<00:00, 3.79it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.36it/s]			
		all	245	482	0.389	0.244
0.214	0.0757					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	54/100	5.02G	1.661	1.818	1.69	10
	100% ██████████		62/62 [00:16<00:00, 3.81it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.34it/s]			
		all	245	482	0.321	0.28
0.198	0.0704					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	55/100	5.02G	1.635	1.729	1.683	15
	100% ██████████		62/62 [00:16<00:00, 3.79it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.35it/s]			
		all	245	482	0.3	0.265
0.198	0.0739					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	56/100	5.02G	1.61	1.705	1.653	11
640:	100% ██████████		62/62 [00:16<00:00, 3.80it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.43it/s]			
		all	245	482	0.339	0.254
0.194	0.0706					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	57/100	5.02G	1.605	1.679	1.646	19
640:	100% ██████████		62/62 [00:16<00:00, 3.79it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.32it/s]			
		all	245	482	0.352	0.269
0.216	0.0806					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	58/100	5.02G	1.614	1.715	1.677	11
640:	100% ██████████		62/62 [00:16<00:00, 3.80it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.40it/s]			
		all	245	482	0.372	0.259
0.217	0.0781					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	59/100	5.02G	1.577	1.651	1.624	11
640:	100% ██████████		62/62 [00:16<00:00, 3.81it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100% ██████████		8/8 [00:01<00:00, 4.43it/s]			

```
0.233      0.0778      all      245      482      0.414      0.268
```

```
Size      Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances
```

```
60/100      5.02G      1.592      1.665      1.632      12
640: 100%|██████████| 62/62 [00:16<00:00, 3.80it/s]
```

```
Class      Images      Instances      Box(P      R
mAP50      mAP50-95): 100%|██████████| 8/8 [00:01<00:00, 4.41it/s]
```

```
0.209      0.0711      all      245      482      0.348      0.268
```

```
Size      Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances
```

```
61/100      5.02G      1.576      1.654      1.635      15
640: 100%|██████████| 62/62 [00:16<00:00, 3.79it/s]
```

```
Class      Images      Instances      Box(P      R
mAP50      mAP50-95): 100%|██████████| 8/8 [00:01<00:00, 4.35it/s]
```

```
0.217      0.0801      all      245      482      0.39      0.238
```

```
Size      Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances
```

```
62/100      5.02G      1.556      1.609      1.615      14
640: 100%|██████████| 62/62 [00:16<00:00, 3.79it/s]
```

```
Class      Images      Instances      Box(P      R
mAP50      mAP50-95): 100%|██████████| 8/8 [00:01<00:00, 4.38it/s]
```

```
0.217      0.0766      all      245      482      0.401      0.264
```

EarlyStopping: Training stopped early as no improvement observed in last 20 epochs. Best results observed at epoch 42, best model saved as best.pt.

To update EarlyStopping(patience=20) pass a new patience value, i.e. `patience=300` or use `patience=0` to disable EarlyStopping.

62 epochs completed in 0.322 hours.
 Optimizer stripped from runs/detect/train/weights/last.pt, 22.5MB
 Optimizer stripped from runs/detect/train/weights/best.pt, 22.5MB
 Validating runs/detect/train/weights/best.pt...
 Ultralytics 8.3.88 Python-3.10.12 torch-2.5.1+cu121 CUDA:0 (Tesla T4, 15095MiB)
 Model summary (fused): 72 layers, 11,126,745 parameters, 0 gradients, 28.4 GFLOPs

	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%		8/8 [00:02<00:00,	3.20it/s]	
	all	245	482	0.39	0.259
0.235	0.088				
	First-degree Burn	96	162	0.341	0.105
0.117	0.0331				
	Second-degree Burn	117	244	0.455	0.25
0.228	0.0745				
	Third-degree Burn	51	76	0.372	0.421
0.361	0.156				

/usr/local/lib/python3.10/dist-packages/matplotlib/colors.py:721:
 RuntimeWarning: invalid value encountered in less
 xa[xa < 0] = -1
 /usr/local/lib/python3.10/dist-packages/matplotlib/colors.py:721:
 RuntimeWarning: invalid value encountered in less
 xa[xa < 0] = -1

Speed: 0.2ms preprocess, 3.8ms inference, 0.0ms loss, 1.9ms
 postprocess per image
 Results saved to runs/detect/train
 YOLOv8 Training Completed!

model.tune(data=f"{YOLO_DATASET_PATH}/data.yaml", epochs=50)

Tuner: Initialized Tuner instance with 'tune_dir=runs/detect/train2'
 Tuner: Learn about tuning at
<https://docs.ultralytics.com/guides/hyperparameter-tuning>
 Tuner: Starting iteration 1/10 with hyperparameters: {'lr0': 0.01,
 'lrf': 0.01, 'momentum': 0.937, 'weight_decay': 0.0005,
 'warmup_epochs': 3.0, 'warmup_momentum': 0.8, 'box': 7.5, 'cls': 0.5,
 'dfl': 1.5, 'hsv_h': 0.015, 'hsv_s': 0.7, 'hsv_v': 0.4, 'degrees':
 0.0, 'translate': 0.1, 'scale': 0.5, 'shear': 0.0, 'perspective': 0.0,
 'flipud': 0.0, 'fliplr': 0.5, 'bgr': 0.0, 'mosaic': 1.0, 'mixup': 0.0,
 'copy_paste': 0.0}
 Saved runs/detect/train2/tune_scatter_plots.png
 Saved runs/detect/train2/tune_fitness.png
 Tuner: 1/10 iterations complete (966.22s)

```
Tuner: Results saved to runs/detect/train2
Tuner: Best fitness=0.10964 observed at iteration 1
Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,
'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,
'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,
'val/cls_loss': 2.65734, 'val/df_l_loss': 2.57768, 'fitness': 0.10964}
Tuner: Best fitness model is runs/detect/train3
Tuner: Best fitness hyperparameters are printed below.
```

```
Printing 'runs/detect/train2/best_hyperparameters.yaml'
```

```
lr0: 0.01
lrf: 0.01
momentum: 0.937
weight_decay: 0.0005
warmup_epochs: 3.0
warmup_momentum: 0.8
box: 7.5
cls: 0.5
df_l: 1.5
hsv_h: 0.015
hsv_s: 0.7
hsv_v: 0.4
degrees: 0.0
translate: 0.1
scale: 0.5
shear: 0.0
perspective: 0.0
flipud: 0.0
fliplr: 0.5
bgr: 0.0
mosaic: 1.0
mixup: 0.0
copy_paste: 0.0
```

```
Tuner: Starting iteration 2/10 with hyperparameters: {'lr0': 0.01,
'lr0': 0.00946, 'momentum': 0.92557, 'weight_decay': 0.00053,
'warmup_epochs': 2.90669, 'warmup_momentum': 0.8, 'box': 7.62173,
'cls': 0.50917, 'df_l': 1.47564, 'hsv_h': 0.01571, 'hsv_s': 0.68308,
'hsv_v': 0.4, 'degrees': 0.0, 'translate': 0.10027, 'scale': 0.50025,
'shear': 0.0, 'perspective': 0.0, 'flipud': 0.0, 'fliplr': 0.48357,
'bgr': 0.0, 'mosaic': 0.91786, 'mixup': 0.0, 'copy_paste': 0.0}
Saved runs/detect/train2/tune_scatter_plots.png
Saved runs/detect/train2/tune_fitness.png
```

```
Tuner: 2/10 iterations complete (1930.25s)
Tuner: Results saved to runs/detect/train2
Tuner: Best fitness=0.10964 observed at iteration 1
Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,
'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,
```



```
'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,  
'val/cls_loss': 2.65734, 'val/dfl_loss': 2.57768, 'fitness': 0.10964}  
Tuner: Best fitness model is runs/detect/train3  
Tuner: Best fitness hyperparameters are printed below.
```

```
Printing 'runs/detect/train2/best_hyperparameters.yaml'
```

```
lr0: 0.01  
lrf: 0.01  
momentum: 0.937  
weight_decay: 0.0005  
warmup_epochs: 3.0  
warmup_momentum: 0.8  
box: 7.5  
cls: 0.5  
dfl: 1.5  
hsv_h: 0.015  
hsv_s: 0.7  
hsv_v: 0.4  
degrees: 0.0  
translate: 0.1  
scale: 0.5  
shear: 0.0  
perspective: 0.0  
flipud: 0.0  
fliplr: 0.5  
bgr: 0.0  
mosaic: 1.0  
mixup: 0.0  
copy_paste: 0.0
```

```
Tuner: Starting iteration 3/10 with hyperparameters: {'lr0': 0.0127,  
'lrf': 0.01, 'momentum': 0.94534, 'weight_decay': 0.00046,  
'warmup_epochs': 3.13593, 'warmup_momentum': 0.7488, 'box': 7.5,  
'cls': 0.4588, 'dfl': 1.74634, 'hsv_h': 0.01508, 'hsv_s': 0.68573,  
'hsv_v': 0.44048, 'degrees': 0.0, 'translate': 0.08762, 'scale': 0.5,  
'shear': 0.0, 'perspective': 0.0, 'flipud': 0.0, 'fliplr': 0.59037,  
'bgr': 0.0, 'mosaic': 0.85607, 'mixup': 0.0, 'copy_paste': 0.0}
```

```
Saved runs/detect/train2/tune_scatter_plots.png
```

```
Saved runs/detect/train2/tune_fitness.png
```

```
Tuner: 3/10 iterations complete (2895.73s)
```

```
Tuner: Results saved to runs/detect/train2
```

```
Tuner: Best fitness=0.10964 observed at iteration 1
```

```
Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,  
'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,  
'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,  
'val/cls_loss': 2.65734, 'val/dfl_loss': 2.57768, 'fitness': 0.10964}
```

```
Tuner: Best fitness model is runs/detect/train3
```

```
Tuner: Best fitness hyperparameters are printed below.
```

Printing 'runs/detect/train2/best_hyperparameters.yaml'

```
lr0: 0.01
lrf: 0.01
momentum: 0.937
weight_decay: 0.0005
warmup_epochs: 3.0
warmup_momentum: 0.8
box: 7.5
cls: 0.5
dfl: 1.5
hsv_h: 0.015
hsv_s: 0.7
hsv_v: 0.4
degrees: 0.0
translate: 0.1
scale: 0.5
shear: 0.0
perspective: 0.0
flipud: 0.0
fliplr: 0.5
bgr: 0.0
mosaic: 1.0
mixup: 0.0
copy_paste: 0.0
```

Tuner: Starting iteration 4/10 with hyperparameters: {'lr0': 0.00841, 'lrf': 0.01097, 'momentum': 0.937, 'weight_decay': 0.0005, 'warmup_epochs': 2.43391, 'warmup_momentum': 0.9362, 'box': 7.96714, 'cls': 0.48526, 'dfl': 1.38426, 'hsv_h': 0.01359, 'hsv_s': 0.7, 'hsv_v': 0.4192, 'degrees': 0.0, 'translate': 0.1, 'scale': 0.48376, 'shear': 0.0, 'perspective': 0.0, 'flipud': 0.0, 'fliplr': 0.38008, 'bgr': 0.0, 'mosaic': 1.0, 'mixup': 0.0, 'copy_paste': 0.0}

Saved runs/detect/train2/tune_scatter_plots.png

Saved runs/detect/train2/tune_fitness.png

Tuner: 4/10 iterations complete □ (3860.62s)

Tuner: Results saved to runs/detect/train2

Tuner: Best fitness=0.10964 observed at iteration 1

Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,

'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,

'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,

'val/cls_loss': 2.65734, 'val/dfl_loss': 2.57768, 'fitness': 0.10964}

Tuner: Best fitness model is runs/detect/train3

Tuner: Best fitness hyperparameters are printed below.

Printing 'runs/detect/train2/best_hyperparameters.yaml'

```
lr0: 0.01
```

```
lrf: 0.01
momentum: 0.937
weight_decay: 0.0005
warmup_epochs: 3.0
warmup_momentum: 0.8
box: 7.5
cls: 0.5
dfl: 1.5
hsv_h: 0.015
hsv_s: 0.7
hsv_v: 0.4
degrees: 0.0
translate: 0.1
scale: 0.5
shear: 0.0
perspective: 0.0
flipud: 0.0
fliplr: 0.5
bgr: 0.0
mosaic: 1.0
mixup: 0.0
copy_paste: 0.0
```

```
Tuner: Starting iteration 5/10 with hyperparameters: {'lr0': 0.00964,
'lr': 0.00954, 'momentum': 0.92098, 'weight_decay': 0.0005,
'warmup_epochs': 3.00462, 'warmup_momentum': 0.81061, 'box': 7.75389,
'cls': 0.5, 'dfl': 1.55109, 'hsv_h': 0.01431, 'hsv_s': 0.7, 'hsv_v':
0.40445, 'degrees': 0.0, 'translate': 0.10555, 'scale': 0.51386,
'shear': 0.0, 'perspective': 0.0, 'flipud': 0.0, 'fliplr': 0.44638,
'bgr': 0.0, 'mosaic': 1.0, 'mixup': 0.0, 'copy_paste': 0.0}
```

Saved runs/detect/train2/tune_scatter_plots.png

Saved runs/detect/train2/tune_fitness.png

Tuner: 5/10 iterations complete (4826.52s)

Tuner: Results saved to runs/detect/train2

Tuner: Best fitness=0.10964 observed at iteration 1

Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,
'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,
'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,
'val/cls_loss': 2.65734, 'val/dfl_loss': 2.57768, 'fitness': 0.10964}

Tuner: Best fitness model is runs/detect/train3

Tuner: Best fitness hyperparameters are printed below.

Printing 'runs/detect/train2/best_hyperparameters.yaml'

```
lr0: 0.01
lrf: 0.01
momentum: 0.937
weight_decay: 0.0005
warmup_epochs: 3.0
```

```
warmup_momentum: 0.8
box: 7.5
cls: 0.5
dfl: 1.5
hsv_h: 0.015
hsv_s: 0.7
hsv_v: 0.4
degrees: 0.0
translate: 0.1
scale: 0.5
shear: 0.0
perspective: 0.0
flipud: 0.0
fliplr: 0.5
bgr: 0.0
mosaic: 1.0
mixup: 0.0
copy_paste: 0.0
```

```
Tuner: Starting iteration 6/10 with hyperparameters: {'lr0': 0.01,
'lr1': 0.00852, 'momentum': 0.89282, 'weight_decay': 0.00037,
'warmup_epochs': 3.09364, 'warmup_momentum': 0.6453, 'box': 7.23945,
'cls': 0.58512, 'dfl': 1.5, 'hsv_h': 0.01659, 'hsv_s': 0.69074,
'hsv_v': 0.39466, 'degrees': 0.0, 'translate': 0.12492, 'scale':
0.54783, 'shear': 0.0, 'perspective': 0.0, 'flipud': 0.0, 'fliplr':
0.49174, 'bgr': 0.0, 'mosaic': 0.81142, 'mixup': 0.0, 'copy_paste':
0.0}
```

```
Saved runs/detect/train2/tune_scatter_plots.png
Saved runs/detect/train2/tune_fitness.png
```

```
Tuner: 6/10 iterations complete (5789.73s)
Tuner: Results saved to runs/detect/train2
Tuner: Best fitness=0.10964 observed at iteration 1
Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,
'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,
'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,
'val/cls_loss': 2.65734, 'val/dfl_loss': 2.57768, 'fitness': 0.10964}
Tuner: Best fitness model is runs/detect/train3
Tuner: Best fitness hyperparameters are printed below.
```

```
Printing 'runs/detect/train2/best_hyperparameters.yaml'
```

```
lr0: 0.01
lr1: 0.01
momentum: 0.937
weight_decay: 0.0005
warmup_epochs: 3.0
warmup_momentum: 0.8
box: 7.5
cls: 0.5
```

```
dfc: 1.5
hsv_h: 0.015
hsv_s: 0.7
hsv_v: 0.4
degrees: 0.0
translate: 0.1
scale: 0.5
shear: 0.0
perspective: 0.0
flipud: 0.0
fliplr: 0.5
bgr: 0.0
mosaic: 1.0
mixup: 0.0
copy_paste: 0.0
```

```
Tuner: Starting iteration 7/10 with hyperparameters: {'lr0': 0.01,
'lr1': 0.00975, 'momentum': 0.92911, 'weight_decay': 0.0005,
'warmup_epochs': 2.97143, 'warmup_momentum': 0.79347, 'box': 7.5,
'cls': 0.52014, 'dfc': 1.54098, 'hsv_h': 0.01443, 'hsv_s': 0.71129,
'hsv_v': 0.3962, 'degrees': 0.0, 'translate': 0.10292, 'scale':
0.49884, 'shear': 0.0, 'perspective': 0.0, 'flipud': 0.0, 'fliplr':
0.50637, 'bgr': 0.0, 'mosaic': 1.0, 'mixup': 0.0, 'copy_paste': 0.0}
Saved runs/detect/train2/tune_scatter_plots.png
Saved runs/detect/train2/tune_fitness.png
```

```
Tuner: 7/10 iterations complete □ (6755.30s)
Tuner: Results saved to runs/detect/train2
Tuner: Best fitness=0.10964 observed at iteration 1
Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,
'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,
'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,
'val/cls_loss': 2.65734, 'val/dfc_loss': 2.57768, 'fitness': 0.10964}
Tuner: Best fitness model is runs/detect/train3
Tuner: Best fitness hyperparameters are printed below.
```

```
Printing 'runs/detect/train2/best_hyperparameters.yaml'
```

```
lr0: 0.01
lr1: 0.01
momentum: 0.937
weight_decay: 0.0005
warmup_epochs: 3.0
warmup_momentum: 0.8
box: 7.5
cls: 0.5
dfc: 1.5
hsv_h: 0.015
hsv_s: 0.7
hsv_v: 0.4
```

```
degrees: 0.0
translate: 0.1
scale: 0.5
shear: 0.0
perspective: 0.0
flipud: 0.0
fliplr: 0.5
bgr: 0.0
mosaic: 1.0
mixup: 0.0
copy_paste: 0.0
```

```
Tuner: Starting iteration 8/10 with hyperparameters: {'lr0': 0.00944,
'lr1': 0.00978, 'momentum': 0.91967, 'weight_decay': 0.00054,
'warmup_epochs': 3.01137, 'warmup_momentum': 0.81061, 'box': 7.83434,
'cls': 0.46717, 'dfl': 1.64098, 'hsv_h': 0.01473, 'hsv_s': 0.7,
'hsv_v': 0.40445, 'degrees': 0.0, 'translate': 0.10555, 'scale':
0.51291, 'shear': 0.0, 'perspective': 0.0, 'flipud': 0.0, 'fliplr':
0.4694, 'bgr': 0.0, 'mosaic': 1.0, 'mixup': 0.0, 'copy_paste': 0.0}
Saved runs/detect/train2/tune_scatter_plots.png
Saved runs/detect/train2/tune_fitness.png
```

```
Tuner: 8/10 iterations complete (7720.94s)
Tuner: Results saved to runs/detect/train2
Tuner: Best fitness=0.10964 observed at iteration 1
Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,
'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,
'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,
'val/cls_loss': 2.65734, 'val/dfl_loss': 2.57768, 'fitness': 0.10964}
Tuner: Best fitness model is runs/detect/train3
Tuner: Best fitness hyperparameters are printed below.
```

Printing 'runs/detect/train2/best_hyperparameters.yaml'

```
lr0: 0.01
lr1: 0.01
momentum: 0.937
weight_decay: 0.0005
warmup_epochs: 3.0
warmup_momentum: 0.8
box: 7.5
cls: 0.5
dfl: 1.5
hsv_h: 0.015
hsv_s: 0.7
hsv_v: 0.4
degrees: 0.0
translate: 0.1
scale: 0.5
shear: 0.0
```

```
perspective: 0.0
flipud: 0.0
fliplr: 0.5
bgr: 0.0
mosaic: 1.0
mixup: 0.0
copy_paste: 0.0
```

```
Tuner: Starting iteration 9/10 with hyperparameters: {'lr0': 0.01079,
'lr1': 0.01023, 'momentum': 0.82469, 'weight_decay': 0.0006,
'warmup_epochs': 3.46189, 'warmup_momentum': 0.67752, 'box': 8.52235,
'cls': 0.46642, 'dfl': 1.56149, 'hsv_h': 0.02285, 'hsv_s': 0.77682,
'hsv_v': 0.3734, 'degrees': 0.0, 'translate': 0.09595, 'scale': 0.5,
'shear': 0.0, 'perspective': 0.0, 'flipud': 0.0, 'fliplr': 0.33453,
'bgr': 0.0, 'mosaic': 1.0, 'mixup': 0.0, 'copy_paste': 0.0}
Saved runs/detect/train2/tune_scatter_plots.png
Saved runs/detect/train2/tune_fitness.png
```

```
Tuner: 9/10 iterations complete (8686.27s)
Tuner: Results saved to runs/detect/train2
Tuner: Best fitness=0.10964 observed at iteration 1
Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,
'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,
'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,
'val/cls_loss': 2.65734, 'val/dfl_loss': 2.57768, 'fitness': 0.10964}
Tuner: Best fitness model is runs/detect/train3
Tuner: Best fitness hyperparameters are printed below.
```

```
Printing 'runs/detect/train2/best_hyperparameters.yaml'
```

```
lr0: 0.01
lr1: 0.01
momentum: 0.937
weight_decay: 0.0005
warmup_epochs: 3.0
warmup_momentum: 0.8
box: 7.5
cls: 0.5
dfl: 1.5
hsv_h: 0.015
hsv_s: 0.7
hsv_v: 0.4
degrees: 0.0
translate: 0.1
scale: 0.5
shear: 0.0
perspective: 0.0
flipud: 0.0
fliplr: 0.5
bgr: 0.0
```

```
mosaic: 1.0
mixup: 0.0
copy_paste: 0.0
```

```
Tuner: Starting iteration 10/10 with hyperparameters: {'lr0': 0.00919,
'lr1': 0.01043, 'momentum': 0.92664, 'weight_decay': 0.0005,
'warmup_epochs': 3.5541, 'warmup_momentum': 0.84644, 'box': 8.55955,
'cls': 0.52123, 'dfl': 1.62156, 'hsv_h': 0.0142, 'hsv_s': 0.69644,
'hsv_v': 0.48195, 'degrees': 0.0, 'translate': 0.09521, 'scale':
0.5331, 'shear': 0.0, 'perspective': 0.0, 'flipud': 0.0, 'fliplr':
0.54729, 'bgr': 0.0, 'mosaic': 0.8934, 'mixup': 0.0, 'copy_paste':
0.0}
```

```
Saved runs/detect/train2/tune_scatter_plots.png
```

```
Saved runs/detect/train2/tune_fitness.png
```

```
Tuner: 10/10 iterations complete (9650.96s)
```

```
Tuner: Results saved to runs/detect/train2
```

```
Tuner: Best fitness=0.10964 observed at iteration 1
```

```
Tuner: Best fitness metrics are {'metrics/precision(B)': 0.44854,
'metrics/recall(B)': 0.27321, 'metrics/mAP50(B)': 0.24214,
'metrics/mAP50-95(B)': 0.09492, 'val/box_loss': 2.57088,
'val/cls_loss': 2.65734, 'val/dfl_loss': 2.57768, 'fitness': 0.10964}
```

```
Tuner: Best fitness model is runs/detect/train3
```

```
Tuner: Best fitness hyperparameters are printed below.
```

```
Printing 'runs/detect/train2/best_hyperparameters.yaml'
```

```
lr0: 0.01
lr1: 0.01
momentum: 0.937
weight_decay: 0.0005
warmup_epochs: 3.0
warmup_momentum: 0.8
box: 7.5
cls: 0.5
dfl: 1.5
hsv_h: 0.015
hsv_s: 0.7
hsv_v: 0.4
degrees: 0.0
translate: 0.1
scale: 0.5
shear: 0.0
perspective: 0.0
flipud: 0.0
fliplr: 0.5
bgr: 0.0
mosaic: 1.0
mixup: 0.0
```



```
copy_paste: 0.0
```

```
metrics = model.val()
```

```
Ultralytics 8.3.88 Python-3.10.12 torch-2.5.1+cu121 CUDA:0 (Tesla T4, 15095MiB)
```

```
Model summary (fused): 72 layers, 11,126,745 parameters, 0 gradients, 28.4 GFLOPs
```

```
val: Scanning /kaggle/working/yolo-dataset/labels/val.cache... 245 images, 0 backgrounds, 0 corrupt: 100%|██████████| 245/245 [00:00<?, ? it/s]
```

	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	16/16	[00:03<00:00, 4.52it/s]	
	all	245	482	0.469	0.245
0.234	0.0883				
	First-degree Burn	96	162	0.442	0.0926
0.117	0.0342				
	Second-degree Burn	117	244	0.487	0.233
0.228	0.0752				
	Third-degree Burn	51	76	0.479	0.408
0.356	0.156				

```
/usr/local/lib/python3.10/dist-packages/matplotlib/colors.py:721: RuntimeWarning: invalid value encountered in less
```

```
    xa[xa < 0] = -1
```

```
/usr/local/lib/python3.10/dist-packages/matplotlib/colors.py:721: RuntimeWarning: invalid value encountered in less
```

```
    xa[xa < 0] = -1
```

```
Speed: 1.0ms preprocess, 9.0ms inference, 0.0ms loss, 1.1ms postprocess per image
```

```
Results saved to runs/detect/train13
```

```
"""WORKING_DIR = "/kaggle/working/"
for root, dirs, files in os.walk(WORKING_DIR):
    print(f"\n Directory: {root}")
    for file in files:
        print(f"    {file}")"""
```

```
RESULTS_PATH = "/kaggle/working/runs/detect/train13/"
```

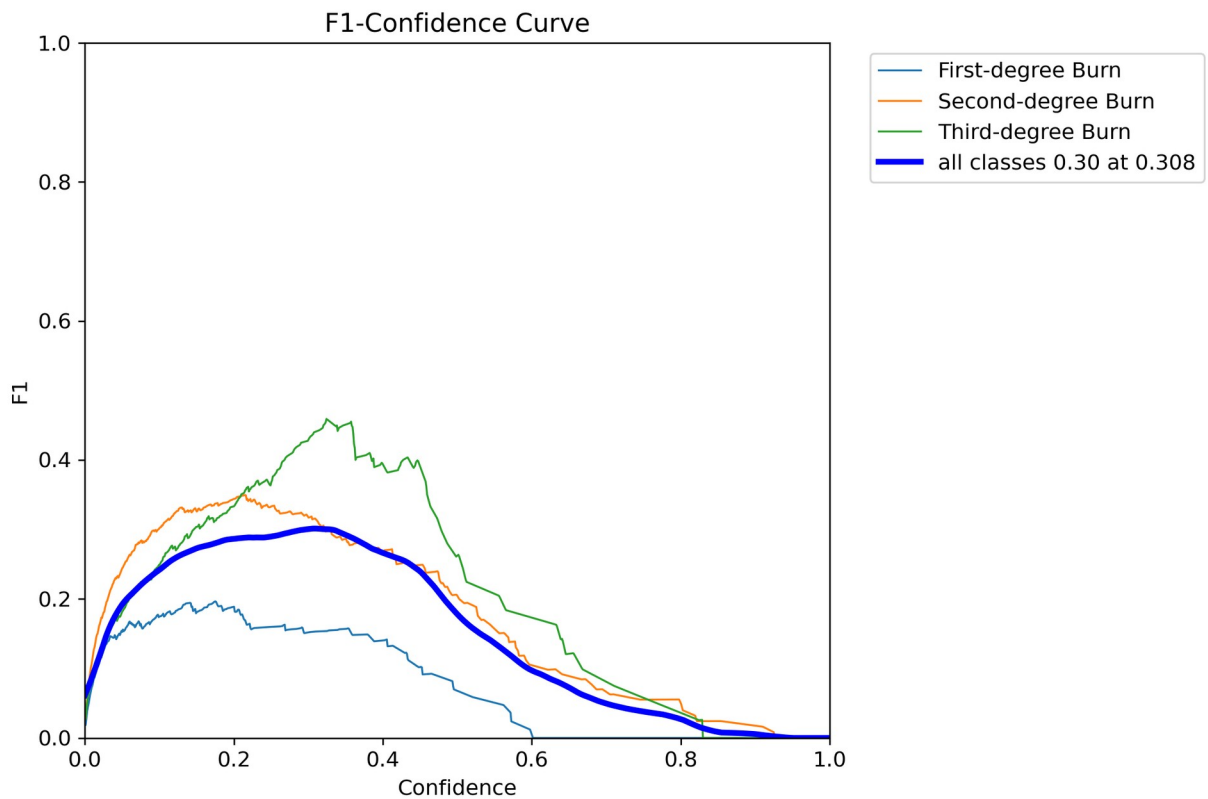
```
plots = [
    "F1_curve.png",
    "PR_curve.png",
    "P_curve.png",
    "R_curve.png",
    "confusion_matrix.png",
    "confusion_matrix_normalized.png",

```

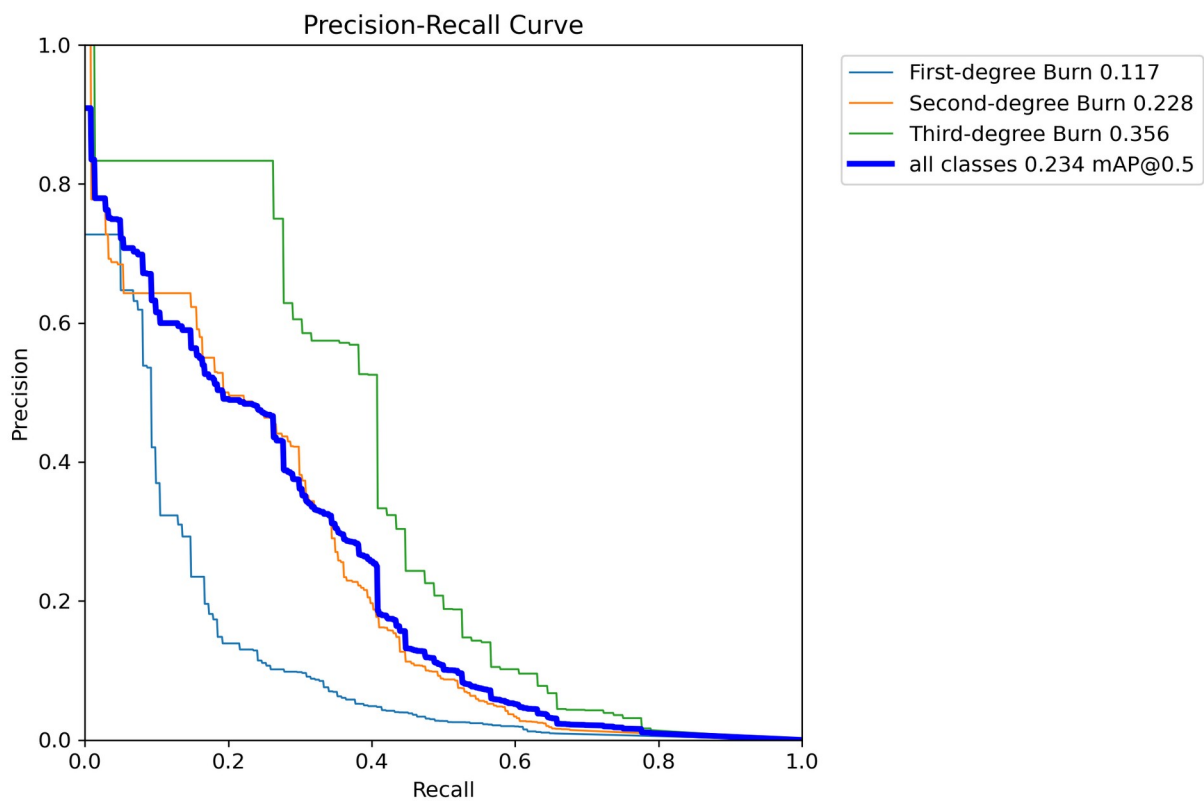
```
]
```

```
for plot in plots:
    plot_path = os.path.join(RESULTS_PATH, plot)
    if os.path.exists(plot_path):
        print(f"Showing {plot}:\n")
        display.display(Image.open(plot_path))
    else:
        print(f"\n{plot} not found!")
```

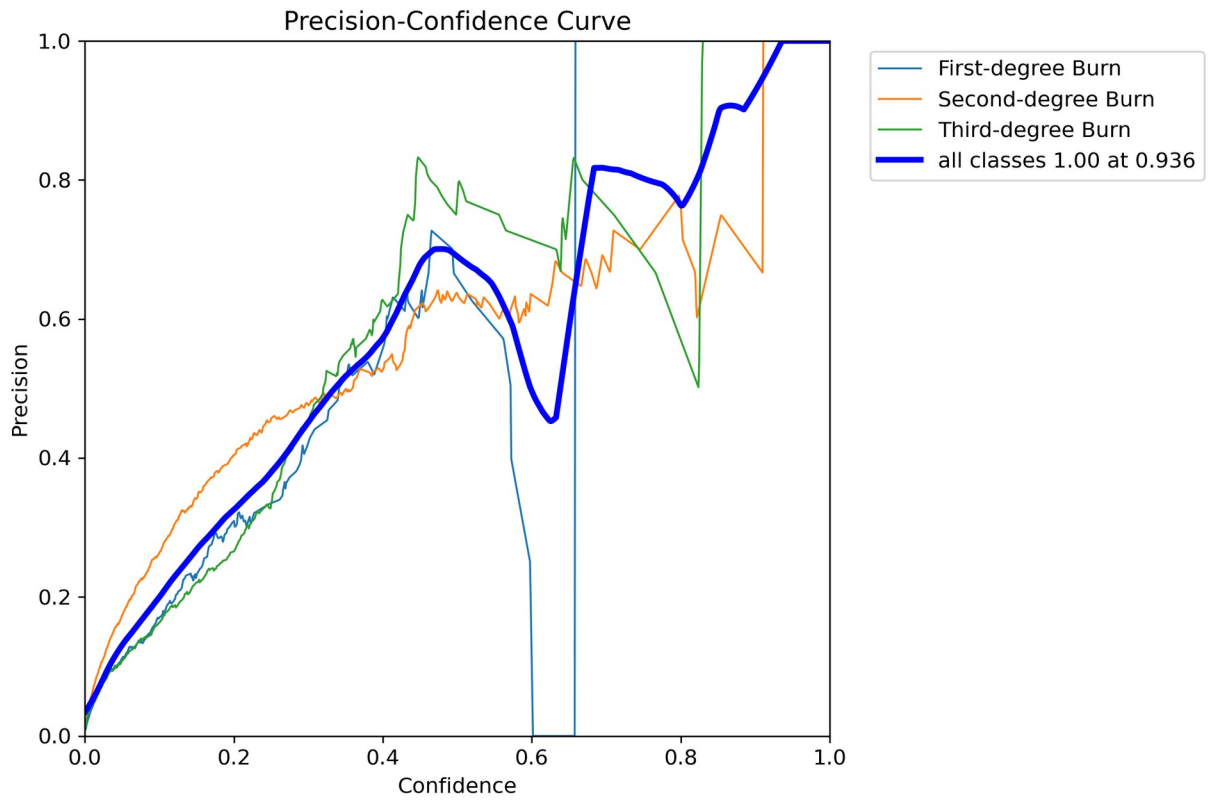
Showing F1_curve.png:



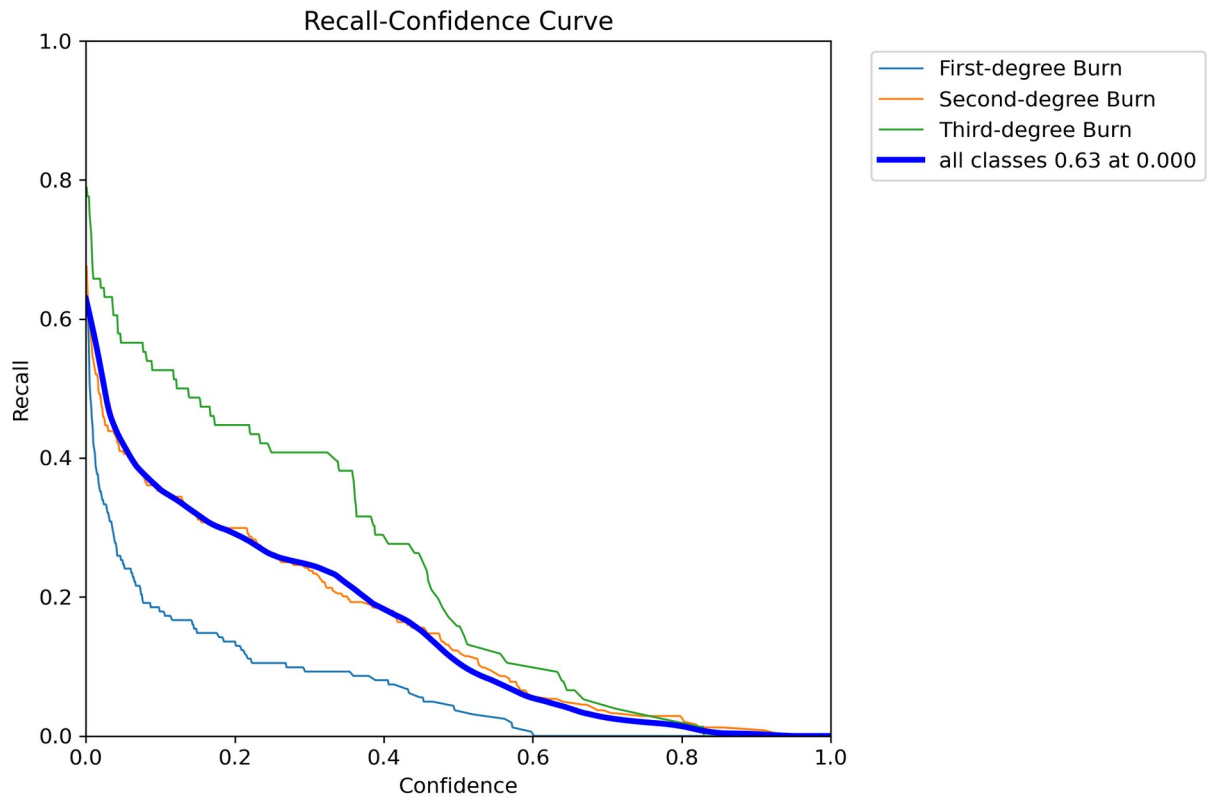
Showing PR_curve.png:



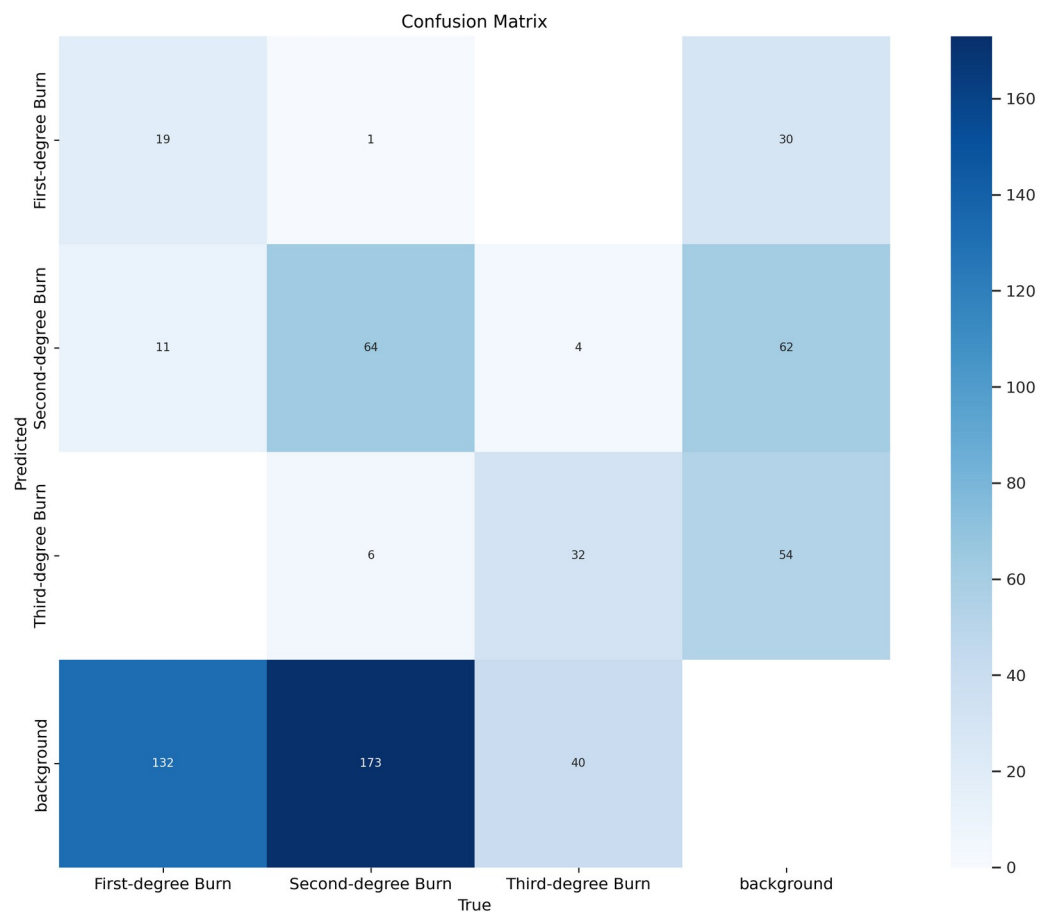
Showing P_curve.png:



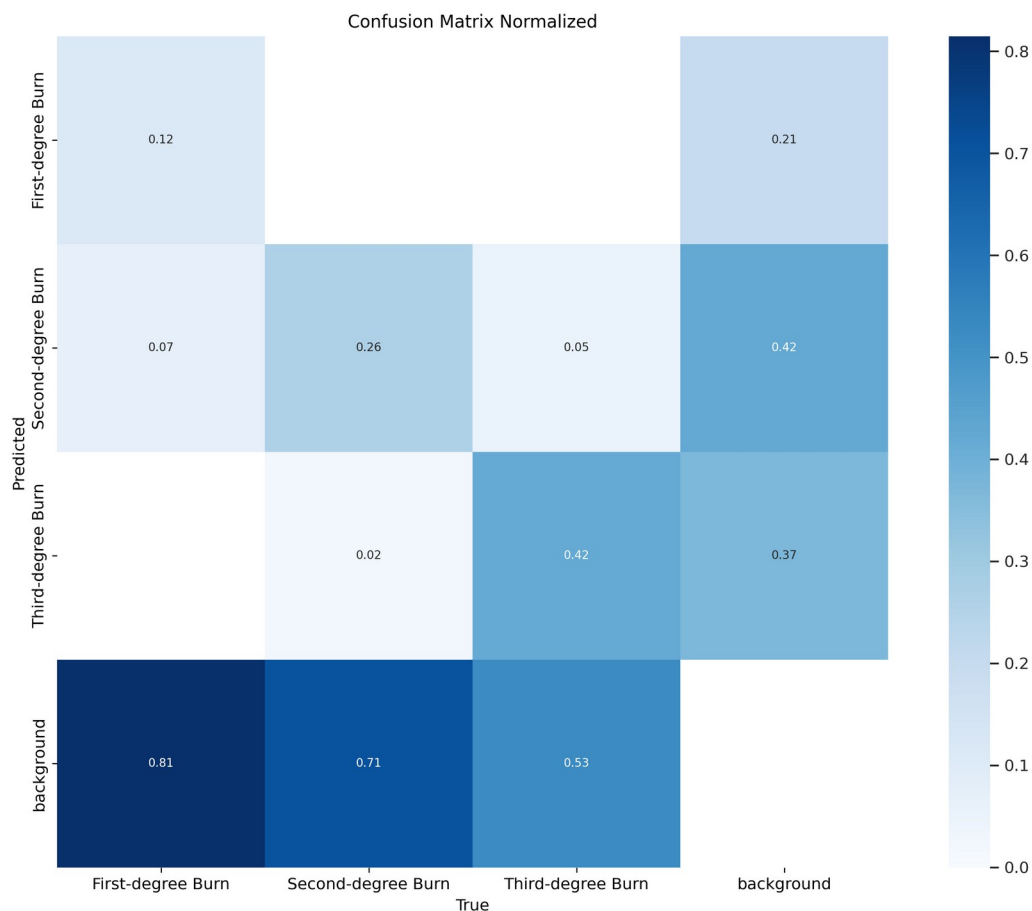
Showing R_curve.png:



Showing confusion_matrix.png:



Showing confusion_matrix_normalized.png:



```
model.export(format="onnx")
best_model_path = "runs/detect/train/weights/best.pt"
shutil.copy(best_model_path, "/kaggle/working/best_yolo_model.pt")
print("Model exported successfully for deployment!")
```

Ultralytics 8.3.88 □ Python-3.10.12 torch-2.5.1+cu121 CPU (Intel Xeon 2.00GHz)

PyTorch: starting from 'runs/detect/train/weights/best.pt' with input shape (1, 3, 640, 640) BCHW and output shape(s) (1, 7, 8400) (21.5 MB)
requirements: Ultralytics requirements ['onnxslim', 'onnxruntime-gpu']
not found, attempting AutoUpdate...

Collecting onnxslim

Downloading onnxslim-0.1.48-py3-none-any.whl.metadata (4.6 kB)

Collecting onnxruntime-gpu

Downloading onnxruntime_gpu-1.21.0-cp310-cp310-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (4.8 kB)

Requirement already satisfied: onnx in /usr/local/lib/python3.10/dist-packages (from onnxslim) (1.17.0)

Requirement already satisfied: sympy in /usr/local/lib/python3.10/dist-packages (from onnxslim) (1.13.1)

Requirement already satisfied: packaging in
/usr/local/lib/python3.10/dist-packages (from onnxslim) (24.2)
Collecting coloredlogs (from onnxruntime-gpu)
 Downloading coloredlogs-15.0.1-py2.py3-none-any.whl.metadata (12 kB)
Requirement already satisfied: flatbuffers in
/usr/local/lib/python3.10/dist-packages (from onnxruntime-gpu)
(24.3.25)
Requirement already satisfied: numpy>=1.21.6 in
/usr/local/lib/python3.10/dist-packages (from onnxruntime-gpu)
(1.26.4)
Requirement already satisfied: protobuf in
/usr/local/lib/python3.10/dist-packages (from onnxruntime-gpu)
(3.20.3)
Requirement already satisfied: mkl_fft in
/usr/local/lib/python3.10/dist-packages (from numpy>=1.21.6-
>onnxruntime-gpu) (1.3.8)
Requirement already satisfied: mkl_random in
/usr/local/lib/python3.10/dist-packages (from numpy>=1.21.6-
>onnxruntime-gpu) (1.2.4)
Requirement already satisfied: mkl_umath in
/usr/local/lib/python3.10/dist-packages (from numpy>=1.21.6-
>onnxruntime-gpu) (0.1.1)
Requirement already satisfied: mkl in /usr/local/lib/python3.10/dist-
packages (from numpy>=1.21.6->onnxruntime-gpu) (2025.0.1)
Requirement already satisfied: tbb4py in
/usr/local/lib/python3.10/dist-packages (from numpy>=1.21.6-
>onnxruntime-gpu) (2022.0.0)
Requirement already satisfied: mkl-service in
/usr/local/lib/python3.10/dist-packages (from numpy>=1.21.6-
>onnxruntime-gpu) (2.4.1)
Collecting humanfriendly>=9.1 (from coloredlogs->onnxruntime-gpu)
 Downloading humanfriendly-10.0-py2.py3-none-any.whl.metadata (9.2
kB)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.10/dist-packages (from sympy->onnxslim) (1.3.0)
Requirement already satisfied: intel-openmp>=2024 in
/usr/local/lib/python3.10/dist-packages (from mkl->numpy>=1.21.6-
>onnxruntime-gpu) (2024.2.0)
Requirement already satisfied: tbb==2022.* in
/usr/local/lib/python3.10/dist-packages (from mkl->numpy>=1.21.6-
>onnxruntime-gpu) (2022.0.0)
Requirement already satisfied: tcmlib==1.* in
/usr/local/lib/python3.10/dist-packages (from tbb==2022.*->mkl-
>numpy>=1.21.6->onnxruntime-gpu) (1.2.0)
Requirement already satisfied: intel-cmplr-lib-rt in
/usr/local/lib/python3.10/dist-packages (from mkl_umath-
>numpy>=1.21.6->onnxruntime-gpu) (2024.2.0)
Requirement already satisfied: intel-cmplr-lib-ur==2024.2.0 in
/usr/local/lib/python3.10/dist-packages (from intel-openmp>=2024->mkl-


```
>numpy>=1.21.6->onnxruntime-gpu) (2024.2.0)
Downloading onnxslim-0.1.48-py3-none-any.whl (142 kB)
_____ 142.9/142.9 kB 5.3 MB/s
eta 0:00:00
Downloading onnxruntime_gpu-1.21.0-cp310-cp310-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (280.8 MB)
_____ 280.8/280.8 MB 212.1 MB/s
eta 0:00:00
Downloading coloredlogs-15.0.1-py2.py3-none-any.whl (46 kB)
_____ 46.0/46.0 kB 187.6 MB/s
eta 0:00:00
Downloading humanfriendly-10.0-py2.py3-none-any.whl (86 kB)
_____ 86.8/86.8 kB 298.1 MB/s
eta 0:00:00
Installing collected packages: humanfriendly, coloredlogs, onnxslim,
onnxruntime-gpu
Successfully installed coloredlogs-15.0.1 humanfriendly-10.0
onnxruntime-gpu-1.21.0 onnxslim-0.1.48

requirements: AutoUpdate success □ 12.4s, installed 2 packages:
['onnxslim', 'onnxruntime-gpu']
requirements: ▲ Restart runtime or rerun command for updates to take
effect

ONNX: starting export with onnx 1.17.0 opset 19...
ONNX: slimming with onnxslim 0.1.48...
ONNX: export success □ 14.2s, saved as
'runs/detect/train/weights/best.onnx' (42.7 MB)

Export complete (16.1s)
Results saved to /kaggle/working/runs/detect/train/weights
Predict:          yolo predict task=detect
model=runs/detect/train/weights/best.onnx imgsz=640
Validate:         yolo val task=detect
model=runs/detect/train/weights/best.onnx imgsz=640
data=/kaggle/working/yolo-dataset/data.yaml
Visualize:        https://netron.app
Model exported successfully for deployment!
```